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Enhancing Operational Efficiency and Energy Savings Through a Fail-Less Artificial Lift Strategy: A Case Study of Long Stroke Unit Deployment in Field A, Petroleum Development Oman

G. Al Shukaili, S. Al Kindi, H. Rady, F. Al Hatmi, A. AlAjmi, Y. AlRiyami, M. Al Lawati, A. Al Ghadani, M. Al Sawafi, and S. Al Matroushi, Petroleum Development Oman, Muscat, Sultanate of Oman

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Abstract

Field A, the largest cluster in Petroleum Development Oman (PDO), faced a significant challenge with a high overall artificial lift failure index, surpassing industry standards at 28% in 2020. The primary contributors to this index were beam pump (BP) wells. In response, a comprehensive workshop was conducted involving all stakeholders to address the issue of artificial lift failures. Field A's artificial lift fail-less strategy was formulated and implemented as a result of this workshop. A long stroke unit (LSU) was deployed in Field A as a result of the persistently high failure rates for conventional beam pump units (BP), which exceeded 32% in 2020. In an effort to mitigate these challenges, a series of successful trials were conducted to evaluate the efficacy of LSU, which led to the conversion of 80 conventional BP units to LSUs between 2020 and 2024.

The results obtained from the deployment of LSU in Field A showcased significantly improved run life, indicated by a reduction in failure index. This increase in reliability contributed to the avoidance of subsurface deferments due to less failures and tripping, resulting in estimated additional production of 320 bbl/d.

Additionally, LSUs reduced the number of well interventions, primarily due to fewer failures. It resulted in a substantial improvement in the life cycle of the wells, saving $2.94 million annually.

As a result of the deployment of LSUs, power savings were also achieved. In comparison to conventional BP units, LSU demonstrated a remarkable 40% reduction in power consumption, resulting in a significant decrease in carbon dioxide equivalent emissions by 4575 tCO2e since LSUs deployment.

This successful implementation not only addressed high failure rates but also enhanced system reliability, reduced corrective maintenance, lowered power consumption, and effectively reduced oil deferment. This abstract outlines these positive outcomes, the challenges faced during implementation, and strategies used to overcome them. Additionally, it presents a roadmap for deploying LSUs across Field ‘A, with plans for over 150 LSUs installation in the next five years. The study provides valuable insights for the oil and gas industry, showcasing the potential for wide adoption of LSUs to enhance performance in similar fields.

Background

Artificial lift is a critical technology in the oil and gas industry, used to enhance the extraction of hydrocarbons from reservoirs, particularly as natural reservoir pressures decline over time. The primary purpose of artificial lift is to lower the bottomhole pressure (BHP), allowing for an increase in production rates by either initiating or enhancing the flow of fluids (oil, gas, or water) to the surface.

In the early stages of a well's life, natural reservoir pressure might be sufficient to push hydrocarbons to the surface. However, as production continues, this pressure decreases, and the natural flow rate decreases. When the natural reservoir drive is no longer enough to lift fluids to the surface, artificial lift systems are employed to sustain or boost production.

Several artificial lift methods have been developed to meet the diverse needs of oil and gas wells. Beam pumps (BPs), or sucker rod pumps, are one of the oldest and most common methods, involving a surface-driven pump connected to a downhole plunger that lifts fluids to the surface, and are widely used in onshore wells. Electrical submersible pumps (ESPs) are high-volume centrifugal pumps placed downhole and powered by electricity, making them ideal for high-production wells, particularly in deep and offshore environments. Progressive cavity pumps (PCPs) are positive displacement pumps that are effective for handling viscous fluids and wells with high solids content, often utilized in heavy oil production. Gas lift involves injecting gas into the wellbore to reduce fluid density, allowing reservoir pressure to more easily push fluids to the surface, and is flexible enough to adapt to varying production conditions.

The selection of an artificial lift method depends on various factors, including the well's depth, production rate, fluid characteristics, and the economic considerations of the field. As the industry continues to innovate, the efficiency and adaptability of artificial lift systems improve, enabling better management of declining reservoir pressures and extending the productive life of oil and gas wells. Artificial lift systems are indispensable for maintaining production in mature fields and optimizing the recovery of hydrocarbons, ensuring that wells continue to produce economically viable quantities of oil and gas even as natural pressures decline. The criteria for selecting the appropriate artificial lift system in the oil industry are summarized in Fig. 1 below.

Artificial Lift Systems Application Screening Values

	Gas Lift	Foam Lift	Plunger Lift	Rod Lift	PCP	ESP	Jet Pump
Max Depth	18,000 ft	22,000 ft	19,000 ft	16,000 ft	8,600 ft	15,000 ft	20,000 ft
	5,486 m	*6,705 m*	*5,791 m*	*4,878 m*	*2,621 m*	*4,572 m*	*6,100 m*
Max Volume	75,000 bpd	500 bpd	200 bpd	6,000 bpd	5,000 bpd	60,000 bpd	35,000
	12,000 m^3/D	*80 m^3/D*	*32 m^3/D*	*950 m^3/D*	*790 m^3/D*	*9,500 m^3/D*	*5,560 m^3/D*
Max Temp	450°F	400°F	550°F	550°F	250°F	482°F	550°F
	232°C	*204°C*	*288°C*	*288°C*	*150°C*	*250°C*	*288°C*
Corrosion Handling	Good to excellent	Excellent	Excellent	Good to Excellent	Fair	Good	Excellent
Gas Handling	Excellent	Excellent	Excellent	Fair to good	Good	Fair	Good
Solids Handling	Good	Good	Fair	Fair to good	Excellent	sand<40ppm	Good
Fluid Gravity (°API)	>15°	>8°	>15°	>8°	8°<API<40°	>15° *Viscosity <400cp*	≥6°
Servicing	Wireline or workover rig	Capillary unit	Wellhead catcher or	Workover or pulling rig	Wireline or workover rig	Wireline or workover rig	N/A
Prime Mover	Compressor	Well natural energy	Well natural energy	Gas or electric	Gas or electric	Electric	N/A
Offshore	Excellent	Good	N/A	Limited	Limited	Excellent	Excellent
System Efficiency	10% to 30%	N/A	N/A	45% to 60%	50% to 75%	35% to 60%	10% to 30%

Excellent	Very Good	Good	Fair	Poor

Figure 1—Artificial lift Systems Application Screening Value

Long Stroke Units (LSUs) represent a significant technological advancement in artificial lift systems, particularly for oil production. While conventional Beam Pumps (BPs) have long been the backbone of artificial lift, providing the necessary force to lift fluids from wells to the surface, their limitations become increasingly evident as oil fields mature and operating conditions become more challenging. Conventional BPs use shorter strokes and higher frequencies, leading to more wear, higher energy use, and shorter run life, especially in challenging wells. LSUs, with their extended stroke lengths, lift fluids more efficiently with fewer strokes, reducing mechanical stress, improving energy efficiency, and extending run life while lowering operational costs. The ability of LSUs to handle difficult conditions, such as high-viscosity fluids or wells with significant depth, makes them a more reliable and cost-effective option in mature fields where conventional BP units struggle to maintain efficiency and reliability. By offering enhanced performance in these challenging environments, LSUs provide a more sustainable solution for maintaining production levels while minimizing downtime and energy consumption. Fig.2 below illustrates the surface units of both a conventional Beam Pump (BP) and a Long Stroke Unit (LSU).

Figure 2—Conventional BP vs. Long Stroke Unit

The primary distinction between LSUs and conventional Beam Pumps lies in the stroke length. While traditional BP units typically operate with stroke lengths of up to 216 inches, LSUs can extend the stroke length to 366 inches or more, depending on the design and application. This extended stroke length offers several key advantages:

- Improved Pump Efficiency: The longer stroke of LSUs reduces the frequency of cycling, which in turn reduces wear and tear on critical components such as sucker rods and pump barrels. This leads to a more stable and efficient pumping operation, especially in wells with high production rates.
- Enhanced Fluid Handling: LSUs are particularly effective in wells with high viscosity fluids or significant amounts of sand production. The longer stroke allows for a smoother fluid flow, reducing the likelihood of pump-off conditions and mechanical failures caused by sand abrasion.
- Energy Efficiency: The design of LSUs enables a more balanced load distribution across the pump's cycle, which can significantly reduce the energy required to lift fluids. Studies have shown that LSUs can reduce energy consumption by up to 40% compared to conventional BP units, making them an attractive option for operators seeking to lower operational costs and carbon emissions.

The deployment of LSUs in various oil fields has demonstrated their effectiveness in enhancing production and reducing failure rates. For instance, a case study conducted in a North American oil field highlighted the ability of LSUs to increase the run life of wells by up to 50%, significantly reducing the frequency of well interventions and associated downtime. This extended run life is critical in fields where high intervention costs and deferred production are major concerns.

Another notable application of LSUs is in mature fields with declining reservoir pressures. In such scenarios, the increased stroke length and efficiency of LSUs can help maintain production levels by optimizing the lift capacity under low-pressure conditions. The ability of LSUs to handle variable operating conditions makes them particularly suitable for fields with fluctuating reservoir characteristics.

Despite their advantages, LSUs are not without challenges. The initial capital investment for LSUs is generally higher than that for conventional BP units, which can be a barrier for some operators. Moreover, LSUs face limitations in certain well conditions, such as severe sand production and complex well geometry. Additionally, the installation of LSUs may require modifications to the existing well foundation, depending on the cellar dimensions, which can further increase costs.

Moreover, the longer stroke length and increased mechanical complexity of LSUs require a higher level of technical expertise for maintenance and operation. This can pose a challenge in fields where technical resources are limited. However, ongoing advancements in LSU technology, including the development of more user-friendly designs and automated monitoring systems, are helping to mitigate these challenges.

Field' A Fail-less strategy

Overview

Artificial lift is a method used to lower the producing bottomhole pressure (BHP) on the formation to obtain a higher production rate from the well. In Field 'A, currently 100% of 2400 oil producer wells are artificially lifted which produce approximately 15,000 m3 of oil per day. The number of artificially lifted wells will continue to increase by 150 wells per year from new oil wells drilling activities. The main types of artificial lift systems used in Field 'A are Progressive Cavity Pumps (PCP) 41%, Beam Pump (BP) 38%, and Electric Submersible Pumps (ESPs) 22%. Failure index of artificial lift in 2020 in Field 'A were 32% for BP, 25% for PCP and 11 % for ESP % associated with an average 3145 bbl/d of subsurface unscheduled deferment.

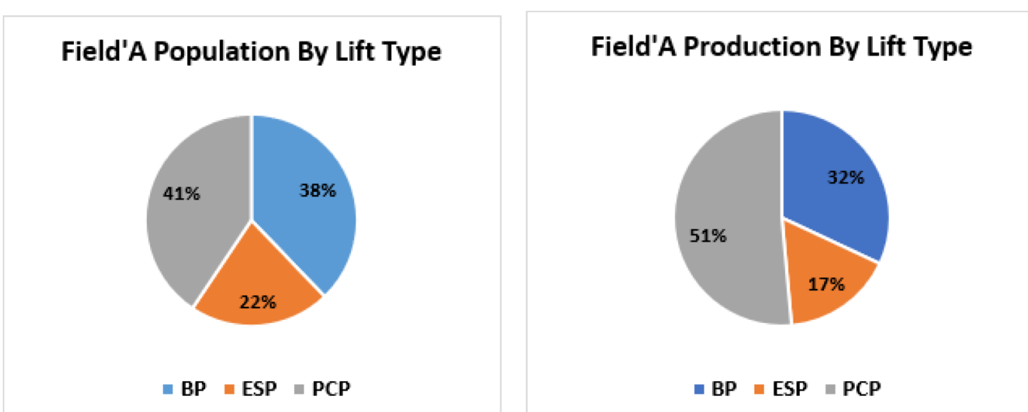

Figure 3—Field 'A Artificial Lift

A large number of resources are utilized in operating, optimizing, maintaining and intervening the artificially lifted wells. Eight hoists and three flush by units(FBU)/ Rapid Service rig (RSR) intervened 1000 wells annually. The majority of them are used to attend the loss of potential (LOP) wells and replace low efficiency pumps (PPR).

Several improvement initiatives were implemented to enhance artificial lift run life, starting with the artificial lift fail-less strategy introduced in 2020. The main recommendations of this strategy included the "Fast Track" conversion of all frequent failure (FF) wells by switching to a different type of artificial lift with better run life, converting "non-FF" wells with good EUR to ESP to improve run life and increase oil production, and aggressively installing LSUs with a target of 30-40 wells per year. Additionally, the strategy emphasized the implementation of AL EPST (enhanced problem-solving technique), a focused effort on reducing artificial lift failures through goal deployment, and the trial and implementation of new artificial lift technologies. Table 1 below presents the 2020 Field ‘A’ Initial Conversion Plan.

Table 1—2020 Field ‘A Initial Conversion Plan

	2020	2021	2022	2023	2024
Conversion to ESP/PCP	35	80	80	35	35
Conversion to LSUs	8	40	40	40	30
Total	**43**	**120**	**120**	**75**	**75**

The artificial lift fail-less strategy is expanded to the new wells by implementing learning from the NFA wells. The objective is to have an artificial lift that is fit for purpose for the full duration of the well's life to obtain a lower capex/opex, lower UTC and improve recovery.

Field ‘A LSUs Trial and Full-Scale Deployment

The deployment of Long Stroke Units (LSUs) in PDO began with a trial phase in 2010, leading to the company's decision to deploy LSUs in 2016. A specific trial phase for instaling LSUs in Field 'A' was conducted in 2019, during which their performance was closely monitored. The effectiveness of LSUs in Field 'A' and similar fields was assessed using several key performance indicators (KPIs), including:

- Run Life (Failure Index): The duration for which the LSU operated without failure, compared to the historical run life of the BP units in the same wells.
- Production Rates: The average daily oil production during the trial period, compared to the pre-conversion production rates.
- Oil Deferment: The average daily deferment during the trial period, compared to the pre-conversion deferment rates for the same wells.
- Power Consumption: The total power consumed by the LSU during the trial period, compared to the power consumption of the BP units.

The LSU trial on Well-X demonstrated a significant improvement in energy efficiency, with the average power usage of conventional pumping units decreasing from 25.6 kW/h to 12.2 kW/h after the installation of the LSU (RF900), marking a 48% reduction in power consumption. While the run life of Well-X was effectively doubled, oil production remained constant; however, the deferment was reduced by 20%.

The trial results were highly promising, with LSUs demonstrating significantly improved run life, reduced failure rates, and lower power consumption.

Based on these results, the Field ‘A’ team decided to move forward with full-scale deployment, successfully converting a total of 70 wells from conventional BP to LSUs since 2020. Building on the success of the LSU deployment in Field A, The Team of Field ‘A has outlined an ambitious plan to further expand the use of LSUs across its operations. Over the next five years, more than 150 additional wells are slated for conversion to LSUs as illustrated in Fig.4 below. In addition, The team formalized a selection criteria to choose the wells for the LSU deployment, ensuring that the wells selected for conversion would provide the most benefits.

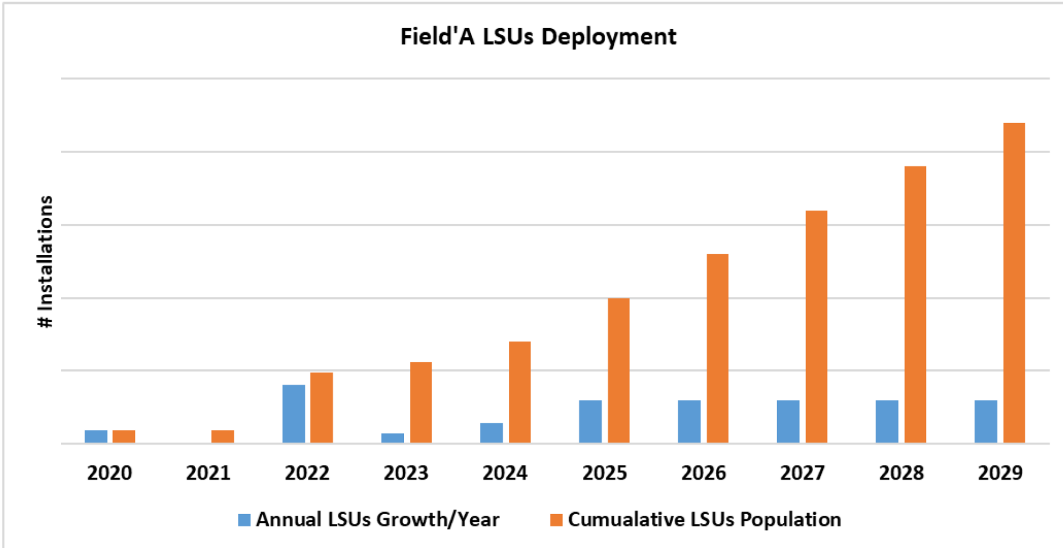

Figure 4—Field ‘A LSU Deployment

Results

The results of the LSU deployment in Field A were analyzed in detail to assess the impact of the technology on operational efficiency, production performance, and environmental sustainability. This section presents a comprehensive analysis of the key outcomes observed following the conversion of BP units to LSUs.

Improved Run Life

One of the most significant outcomes of the LSU deployment was the dramatic improvement in run life observed across the converted wells. The average run life of wells equipped with LSUs increased by 50%, from an average of 12 months with BP units to 18 months with LSUs as demonstrated in Fig.5 below. This improvement was attributed to the more robust design of LSUs, which are better suited to handle the mechanical stresses and challenging operating conditions of Field A.

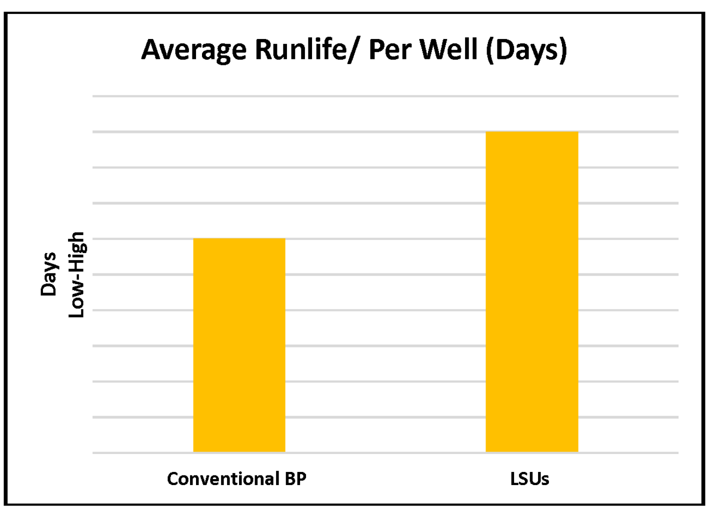

Figure 5—Average Run life of LSU Vs BP

Fig.6 shows the reduction in failure index in BP wells (conventional + LSUs) in Filed ‘A. The failure index of BP wells in Field ‘A for dropped from 32% in 2020 to 24% in 2024. This reduction in failures had a direct impact on well uptime, as fewer failures meant fewer well shutdowns and workover operations, hence less operating cost(OPEX).

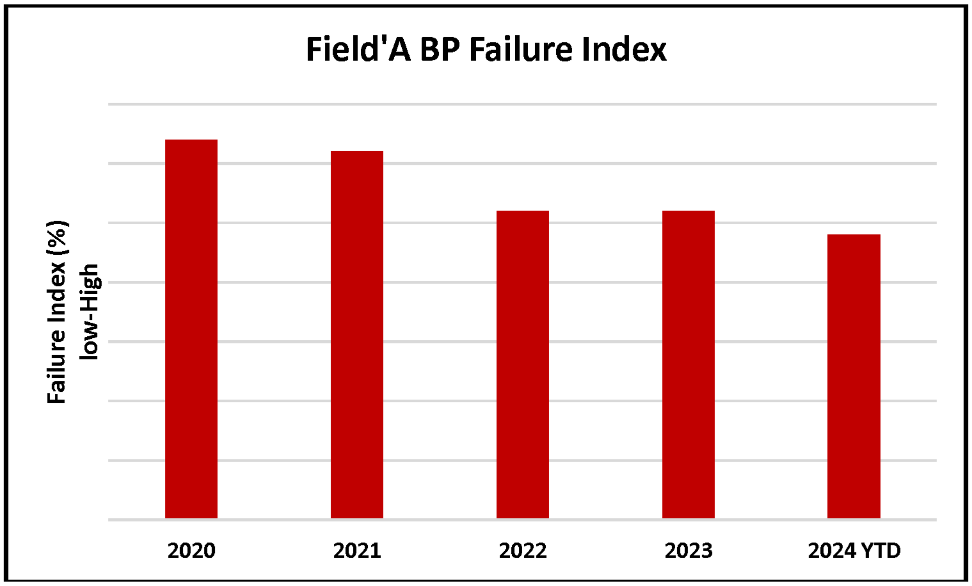

Figure 6—Field ‘A BP Failure Index

Enhanced Production Performance

The improved run life and reduced failures of LSUs had a positive impact on overall production performance in Field A. The average daily production from wells equipped with LSUs increased by 20%, compared to the pre-conversion production rates as can be seen in Fig.7 below. This increase was primarily due to relatively higher gross production as result of deeper installation. This increase was primarily due to the reduced downtime and more consistent operation of the wells, which allowed for more stable and sustained production.

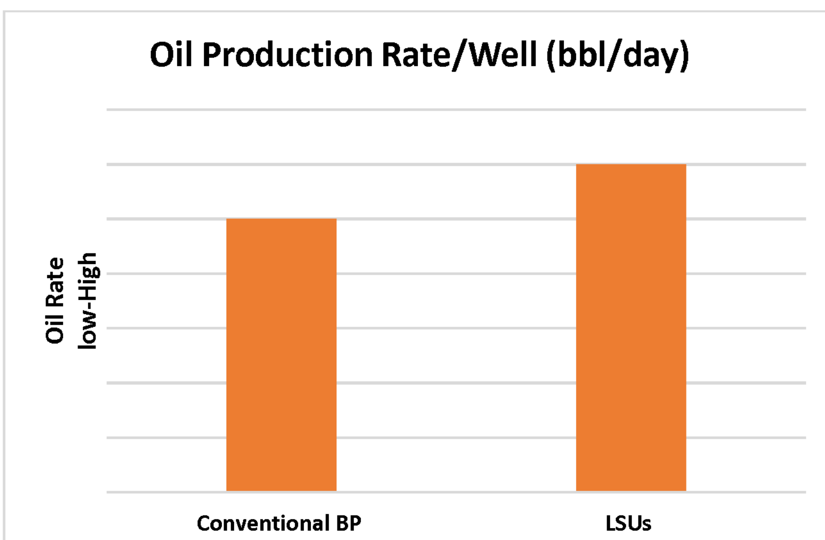

Figure 7—Oil Production Rate LSU vs. BP

The improved reliability of LSUs also contributed to a significant reduction in deferred production. During the past deployment period, it is estimated that the use of LSUs avoided approximately 320 bbl/d of deferred production, which would have been lost due to well failures and associated downtime.

The deferment rate for wells equipped with LSUs was reduced by nearly 50% compared to conventional BPs, as shown in Fig.8 below. This improvement is attributed to decreased downtime and more consistent well operation, leading to more stable and sustained production.

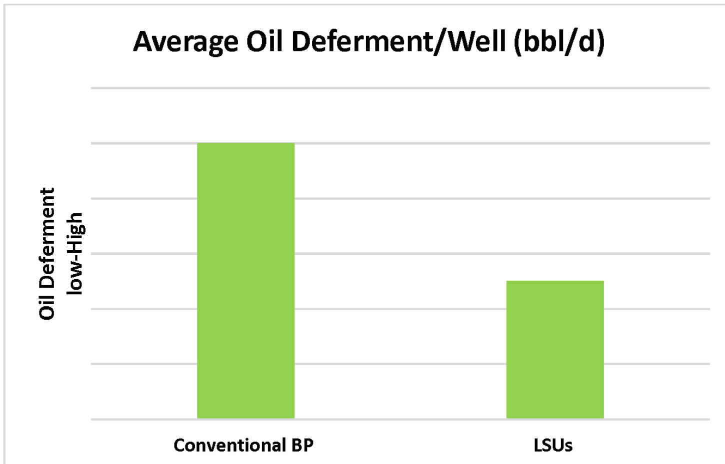

Figure 8—Deferment LSU vs BP

Power Savings and Environmental Benefits

Another key outcome of the LSU deployment was the significant reduction in power consumption observed across the converted wells. Fig.9 shows the comparison of average power consumption (in MW) between conventional Beam Pumps (BP) and Long Stroke Units (LSU). On average, wells equipped with LSUs consumed 40% less power compared to those equipped with conventional BP units. This reduction in power consumption was attributed to the more efficient operation of LSUs, which require less energy to lift the same volume of fluid due to their longer stroke length and optimized pumping dynamics. The power savings achieved through the use of LSUs had a direct impact on the field's carbon footprint. It was measured the average saving per one LSU installation relative to conventional BP is 0.007 MW.

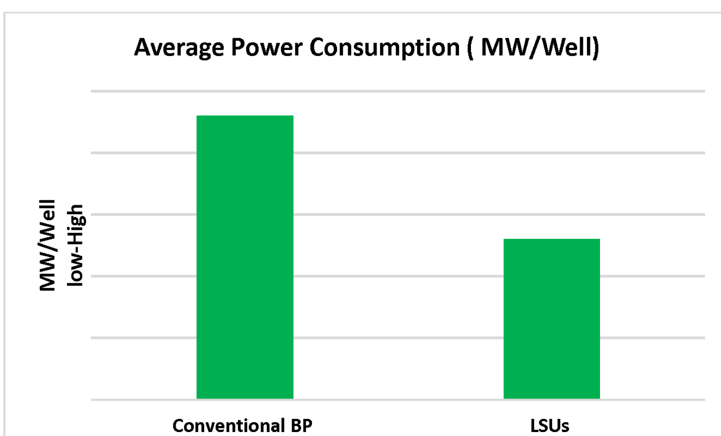

Figure 9—Average Power Consumption

It is estimated that the deployment of LSUs in Field A resulted in a cumulative reduction of 4575 tCO2e in carbon dioxide equivalent emissions over 2020-2024 deployment period as illustrated in Fig.10. This reduction in emissions aligns with PDO's broader sustainability goals and demonstrates the environmental benefits of adopting more energy-efficient technologies in oil field operations.

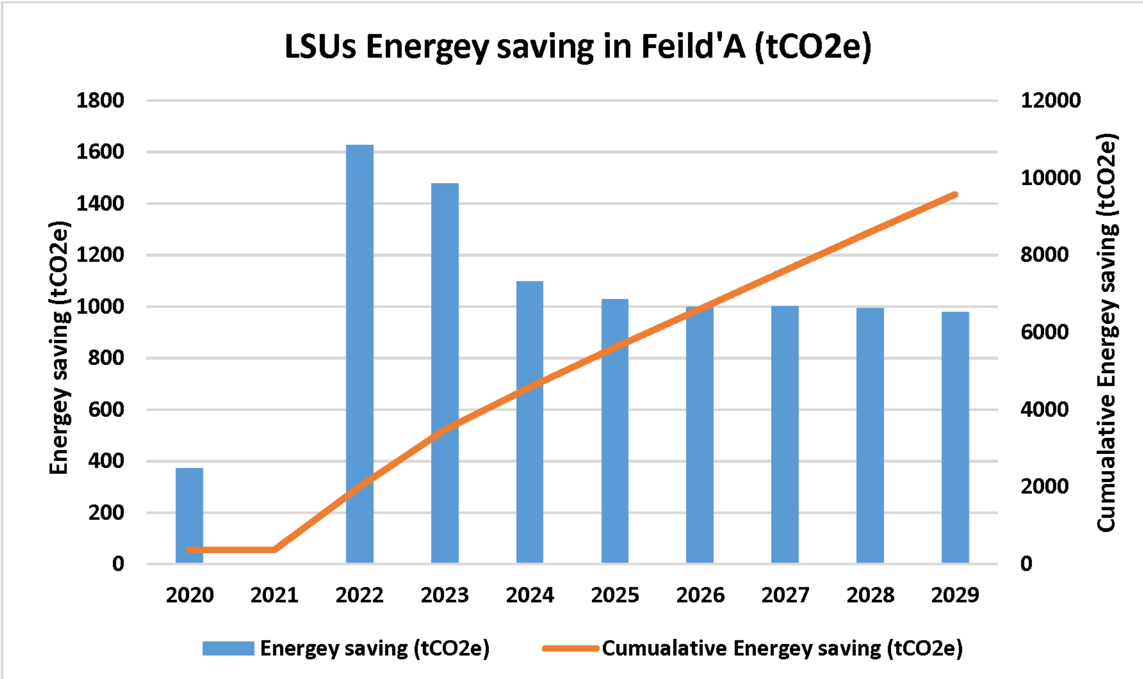

Figure 10—LSUs Energy saving in Field'A (tCO2e)

The environmental benefits of LSUs extend beyond power savings. The improved reliability and reduced failure rates of LSUs also contributed to a decrease in the frequency of well interventions, which are often associated with significant environmental risks, such as oil spills and emissions of gases. By reducing the need for frequent interventions, LSUs helped to minimize these environmental risks, further enhancing the sustainability of Field A's operations.

Cost Saving from well interventions

The average annual cost savings per well for LSUs, compared to conventional beam pumps, is approximately $98,000, resulting in a total yearly operational expenditure (OPEX) savings of $2.94 million annually. These savings are primarily due to the reduced need for well interventions, such as well pulling hoists (WPH) or flash by units (FBU).

Challenges and Limitations

The deployment of LSUs in Field 'A' and PDO faced several challenges. One major hurdle was the qualification of LSU suppliers to ensure they met the company's commercial and technical requirements, which is crucial for supporting the vision of large-scale deployment. The selection of the right suppliers is critical, as it directly influences the success and sustainability of LSU operations. Additionally, delays in material delivery from suppliers led to setbacks in the commissioning process, further impacting the deployment timeline.

Furthermore, significant challenges arose with the well foundations, primarily due to intervention limitations and Health, Safety, and Environmental (HSE) concerns. To overcome these issues, modifications to the well foundations were necessary for some wells before LSU installation. These modifications depended on the cellar dimensions and were required to ensure the successful and safe deployment of the LSUs.

Conclusion

The deployment of Long Stroke Units (LSUs) in Field A has proven to be a transformative initiative, significantly enhancing both operational efficiency and environmental sustainability. The conversion of conventional Beam Pump (BP) units to LSUs resulted in a substantial increase in run life, with a 50% improvement observed across the converted wells. This extended run life directly contributed to a reduction in well failures, which in turn minimized subsurface deferments and boosted production output by approximately 320 bbl/d.

Furthermore, the implementation of LSUs led to impressive power savings, with an average reduction of 40% in energy consumption per well. This efficiency translated into a notable decrease in carbon emissions, aligning with Petroleum Development Oman's (PDO) sustainability goals by reducing the carbon footprint by 4575 tCO2e over the deployment period. The financial impact was equally significant, with cost savings of $2.94 million annually due to the reduced need for well interventions.

The successful deployment of LSUs not only addressed the high failure rates associated with conventional BP units but also demonstrated the potential for wide adoption of this technology across similar fields. The insights gained from this study provide a clear roadmap for future deployments, ensuring that LSUs continue to play a critical role in enhancing the reliability and sustainability of oil production operations.

As PDO looks to expand the use of LSUs across its fields, this project serves as a benchmark for how innovative technologies can drive both operational excellence and environmental stewardship in the oil and gas industry. The lessons learned from Field A's experience offer valuable guidance for operators seeking to overcome similar challenges and achieve sustainable deployment of artificial lift systems.

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